

Homework Section 15.1

1.
 - a) Estimate the volume of the solid that lies below the surface $z = xy$ and above the rectangle $R = \{(x, y) \mid 0 \leq x \leq 6, 0 \leq y \leq 4\}$. Use a Riemann sum with $m = 3$, $n = 2$, and take the sample point to be the upper right corner of each subrectangle.
 - b) Repeat part (a) using midpoints as your sample points.
2.
 - a) Estimate the volume of the solid that lies below the surface $z = x + 2y^2$ and above the rectangle $R = [0, 2] \times [0, 4]$. Use a Riemann sum with ~~$m = 3$~~ , $n = 2$, ~~and~~ $m = 2$ take the sample point to be the ~~upper~~ right corner of each subrectangle.
 - b) Repeat part (a) using ~~lower~~ midpoints as your sample points.
3. A table of values is given for a function $f(x, y)$ defined on $R = [1, 3] \times [0, 4]$.
 - a) Estimate $\iint_R f(x, y) dA$ using midpoints with $m = n = 2$.
 - b) Estimate the double integral with $m = n = 4$ by choosing the sample points to be the points farthest from the origin.

$y \backslash x$	0	1	2	3	4
1	2	0	-3	-6	-5
1.5	3	1	-4	-8	-6
2	4	3	0	-5	-8
2.5	5	5	3	-1	-4
3	7	8	6	3	0

4. Evaluate the double integral by first identifying it as the volume of a solid. ([Example](#))
 - a) $\iint_R 3dA$, $R = \{(x, y) \mid -2 \leq x \leq 2, 1 \leq y \leq 6\}$
 - b) $\iint_R (5 - x)dA$, $R = \{(x, y) \mid 0 \leq x \leq 5, 0 \leq y \leq 3\}$